MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON, LANDBASED ARRESTING GEAR, 11-INCH

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements of an 11-inch nylon tape having a breaking strength of not less than 150,000 pounds.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-13239	Barrier Material, Waterproofed, Flexible, All Temperature
MIL-B-81374	Barrier Material, Greaseproof, Waterproof, Flexible, High Strength

STANDARDS

Federal

Fed. Test Method Std. No. 191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 8315

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DRAWINGS

Naval Air Engineering Center

508306	Reel, Tape Assembly
508327	Purchase Tape Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 25380.)

U.S. Air Force Specification Bulletin No. 90 Color Codes Manufacturers' Identification, Nylon Webbing

(Application for copies should be addressed to the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials -

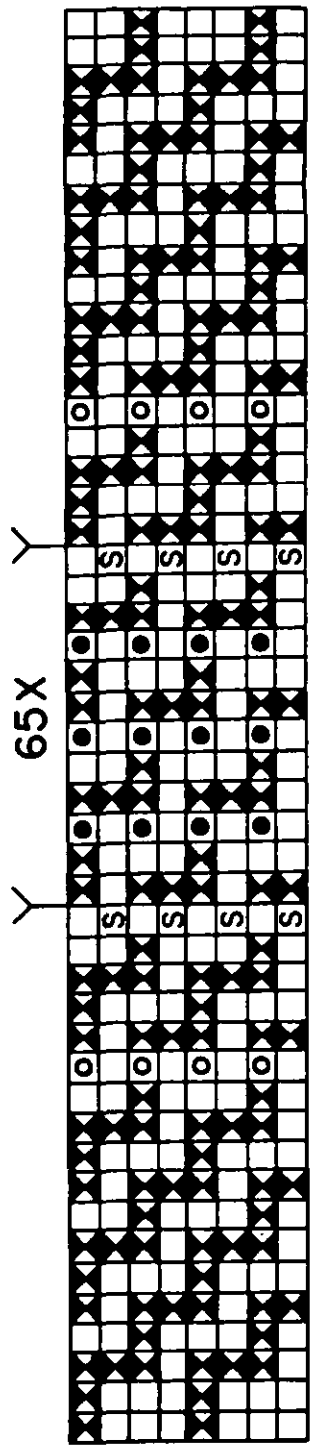
3.1.1 Yarn - The yarn used in the manufacture of the tape shall be 840 denier, 8.5 grams per denier minimum, continuous 140 filament, bright, high-tenacity, light and heat resistant nylon yarn. It shall be polyamide prepared from hexamethylene diamine and adipic acid, or its derivatives, and shall have a minimum melting point of 244° C. Yarn designed for cordage use is not permitted in ground and binder warps. The yarn shall be substantially free from sizing, loading, or other adulterants, and shall not be bleached in any manner or by any process.

3.2 Weave - The weave shall be as shown in Figure 1.

3.3 Color - The color of the tape shall be natural.

3.4 Construction and physical properties - The finished nylon tape shall conform to the construction and physical properties of Table 1.

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KEY TO MARKS

- BODY SINGLY
- STUFFERS 9 AS 1
- STUFFERS 5 AS 1
- BINDER SINGLY 1/

1/ SEE TABLE I FOR ALTERNATE CONSTRUCTION 2 AS 1

FIGURE 1

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TABLE I

PHYSICAL PROPERTIES

CHARACTERISTIC	REQUIREMENT
Width, inches	10.88 to 11.13
Thickness, inches	0.295 ±.010
Weight, ounces/yard (max)	62.0
Breaking strength, pounds (min) 1/	150,000
Ends in warp (min)	
Ground	553
Binder	66 2/
Stuffer	1765
Filling, yarns per inch (min)	13
Yarn ply (min)	
Ground	6
Binder	6 2/
Stuffer	6
Filling	7
Ply twist, turns per inch	
Ground	1.0 - 2.0
Binder	1.0 - 2.0
Stuffer	1.0 - 2.0
Filling	1.0 - 2.0

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1/ A failure at or above 150,000 pounds tension or a jaw break at or above 150,000 pounds will be considered acceptable. A failure below 150,000 pounds tension or a jaw break below 150,000 pounds will be considered unacceptable.

2/ Alternate Construction is 132 ends in warp with 3 yarn ply binder using 2 as 1.

3.5 Length of roll - Unless otherwise specified, the length of each production tape shall be as specified in NAEL(SI) Drawing No. 508327.

3.6 Identification -

3.6.1 Manufacturer's identification - Each manufacturer shall incorporate, as part of binder warp at the center of the tape, two ends dyed to match the shade assigned to that manufacturer in accordance with Air Force Bulletin No. 90. A producer not listed in Bulletin No. 90 shall apply to the procuring activity for assignment of color coding. The colorfastness of this yarn shall be rated as "Fair" or "Good" when tested by method specified in 4.7.

3.7 Fiber identification - Each roll of tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship - The finished tape shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels specified herein.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections - The inspection requirements specified herein are classified as follows:

(a) Inspection of components (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Inspection of components - Quality conformance provisions for components described shall be in accordance with subsidiary specifications and

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drawings referenced to the extent applicable, except that this specification shall govern in the event of conflict. In addition, the supplier shall furnish a certificate of compliance for the characteristics referenced in 3.1.1. The certificate of compliance shall be accompanied by actual test, inspection, or other verified quality data.

4.4 Quality conformance inspection - The tape shall be tested and inspected as noted in 4.5 and must pass the requirements shown in Table II.

4.5 Quality conformance tests - The quality conformance tests shall consist of:

(a) Sampling Plan A and tests

(b) Sampling Plan B and tests

(c) Sampling Plan C and tests

(d) Sampling Plan D and tests

4.6 Description of sampling plans - Tests for each sampling plan are those shown in Table II alongside the appropriate plan letter with the test method as shown therein.

4.6.1 Sampling Plan A - Tests shall be conducted on every production tape length.

4.6.2 Sampling Plan B - Tests shall be conducted on one sample for each continuous loom run.

4.6.3 Sampling Plan C - Tests shall be conducted on samples taken prior to weaving from every 10,000 pounds of yarn or fraction thereof.

4.6.4 Sampling Plan D - For this inspection, sampling shall be conducted in accordance with 4.6.4.1.

4.6.4.1 Inspection levels - The inspection levels for Sampling Plan D shall be as follows:

(a) Yard-by-yard - A one yard unit of tape shall be inspected at intervals of approximately 40 feet of tape as specified in 4.7.4.1 and 4.7.4.1.1.

(b) Overall - The pieces examined shall be those selected yard-by-yard examination.

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4.7 Test methods -

4.7.1 The tests to be run are shown in Table II and applicable paragraphs noted therein. The physical and chemical values specified in Section 3 of this specification apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. Failure of any sample of tape to conform to the requirements specified in Section 3 shall be cause for rejection of the lot represented. Atmospheric conditions for the physical tests shall be in accordance with Fed. Test Method Std. No. 191.

TABLE II

TEST AND TEST METHODS

SAMPLING PLAN	TEST	METHOD	
		FED. TEST METHOD STD. NO. 191	OTHER
A	Width	5020	
A	Thickness	5030 1/	
A	Weight	5040	
A	Length	5010	
A	Breaking strength	——	4.7.2
B	Ends in warp		
	Ground		Visual
	Binder		Visual
	Stuffer		Visual
B	Filling, yarns per inch	5050	
B	Yarn ply		
	Ground		Visual
	Binder		Visual

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TABLE II (Continued)

SAMPLING PLAN	TEST	METHOD	
		FED. TEST METHOD STD. NO. 191	OTHER
B	Yarn ply (Cont'd)		
	Stuffer		Visual
	Filling		Visual
B	Twist, turns per inch		
	Ground	4054 2/	
	Binder	4054 2/	
	Stuffer	4054 2/	
	Filling	4054 2/	
B	Weave		Figure 1
C	Colorfastness to laundering	5614 3/ 4/	
C	Melting point	1534 5/	
C	Denier	4021 5/	
C	Filament count	5/	Visual
D	Classification of defects		
	Yard-by-yard		4.7.4.1
	Overall		4.7.4.2

1/ Except that a six ounce total load shall be applied and the presser foot diameter shall be 3/8 inch.

2/ Five samples each of the ground, binder, stuffer, and filler yarns from woven tape shall be tested.

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3/ Applicable only to manufacturer's identification yarns.

4/ Except that nylon color transfer cloth shall be used.

5/ Unless otherwise specified, a supplier's certificate of compliance is required and will be accepted for the requirement.

4.7.2 Breaking strength - The breaking strength test shall be conducted in accordance with 4.7.2.2. The type and capacity of the tensile testing machine shall be suitable for the type of tape undergoing tests. The machine shall be adjusted so that no-load separation of the grips or pulling jaws shall be at the rate of travel specified herein. All breaking tests shall be made on full width sections of webbing. Two satisfactory sample tests of each three samples tested shall be considered proof of satisfactory tape strength of the specimen.

4.7.2.1 The test grips for holding the specimen shall be so designed that tape failure below 150,000 pounds will not occur within the grip. The no-load rate of jaw separation shall be 4 to 8 inches per minute.

4.7.2.2 Tape ultimate strength tests - A test specimen, 30 feet long, shall be obtained as described in 4.7.2.3. The specimen shall be load tested.

4.7.2.3 Method of sampling for breaking strength tests - The first section of tape leaving a loom run shall have a 30 foot specimen removed from the lead end. After this, a 30 foot specimen shall be removed immediately following every other, or alternate, production tape. The rolls of tape must be marked consecutively as they leave the loom, with the 30 foot specimens marked identically with the rolls of tape to which they apply. Test the first 30 foot specimen leaving the loom run and every 30 foot specimen thereafter. The three 30 foot specimens are tested as outlined in 4.7.2.2. The tests may result in one of the following four possibilities (see Figure 2):

4.7.2.3.1 Test result I - If the two tape specimens pass the strength test, both production tapes in between them are satisfactory.

4.7.2.3.2 Test result II - If the two tape specimens fail the strength test, the production tapes before and after the failed test specimens are unsatisfactory. In this case the first three production tapes are unsatisfactory.

4.7.2.3.3 Test result III - If the first specimen leaving the loom run passes the strength test and the specimen following the second production tape fails, the production tapes before and after the failed specimen are unsatisfactory. The condition of the first production tape must be determined by testing a 30-foot specimen taken from the lead end of the failed second production tape itself, since no provision was made for an additional 30-foot specimen at the end of the first production tape. If the specimen fails, the first production tape is unsatisfactory.

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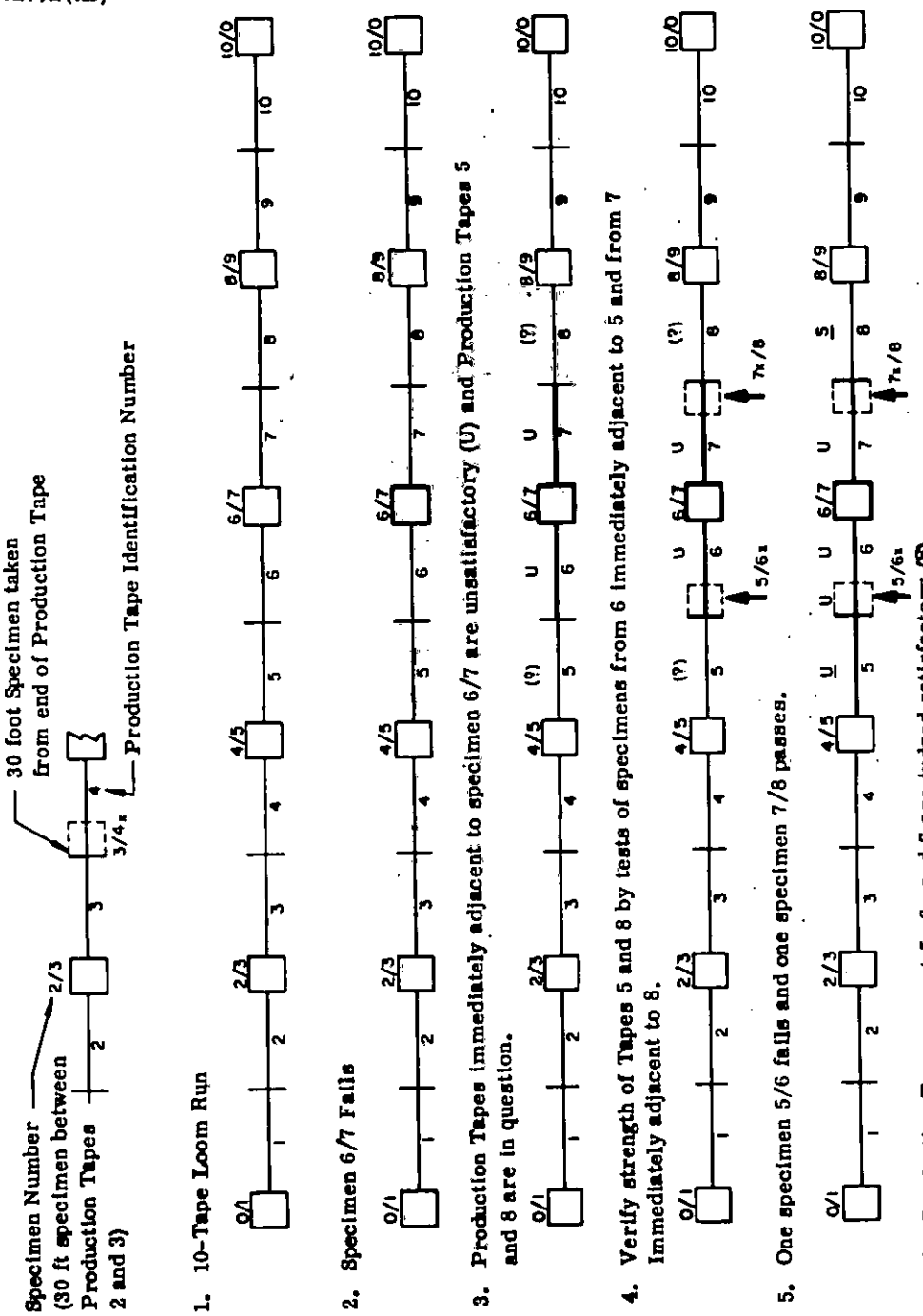


Figure 2.

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4.7.2.3.4 Test result IV - If the first specimen leaving the loom run fails the strength test and the specimen following the second production tape passes, the production tape following the failed specimen is unsatisfactory. The condition of the second production tape must be determined by testing a 30-foot specimen from the trailing end of the failed production tape, since no provision was made for an additional 30 foot specimen at the beginning of the second production tape. If the specimen passes, the second production tape is satisfactory. If the specimen fails, the second production tape is unsatisfactory. Sampling of a longer loom run should be continued by testing each 30-foot test specimen using the procedure outlined for the first and second production tapes. (See Figure 2, which illustrates a hypothetical production loom run sampling procedure for 10 production tapes.)

4.7.3 Denier, filament count, and melting point - An affidavit from the yarn manufacturer will be acceptable as proof that type of nylon, yarn strength, denier, filament count, and melting point are met. If the approximate melting point of the specimen is not known before testing, it shall be determined by a trial run.

4.7.4 Classification of defects - Defects found during the examination shall be classified in accordance with 4.7.4.1 and 4.7.4.2. The Government reserves the right to require examination for any defects not included herein and to classify such defects in accordance with definition contained in MIL-STD-105.

4.7.4.1 Yard-by-yard examination - A one-yard unit of tape as indicated in 4.6.4.1 (a) shall be examined and visual defects classified as listed in Table III. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the tape, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard, or fraction thereof, in which it occurs.

TABLE III

CLASSIFICATION OF DEFECTS

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Abrasion Marks	Resulting in rupture of yarns, or in nap sufficient to obscure the identity of any yarn exceeding 10 percent of width or one inch in length.	X		
Broken or Missing End	Two or more regardless of length of a single end exceeding 6 inches in length.	X		
	Single end under 6 inches but exceeding 1/4 inch.			X

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TABLE III (Continued)

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Broken or Missing Pick	Two or more regardless of extent. NOTE: The filling tie-in or joining shall not be construed as a defect of any nature.	X		
Coarse or Light Filling Bar	Resulting in visible difference in stiffness or thickness of tape and extending for more than 1/4 inch or less in the length direction.	X		
Twist or Distortion	Tape will not lay flat upon application of manual pressure due to twist or distortion.			X
Cut, Hole or Tear	Any cut, hole or tear.	X		
Drop-Ply	Clearly visible on more than two ends within same length and extending over 9 linear inches or more.		X	
Edge Loop	Forming clearly visible filling loops, or edges tied loosely to body of webbing for 2 linear inches or more.	X		
Edge Cut, Torn or Frayed	Clearly visible along edge.	X		
Edge Scalloped	Any indentation of edges above allowable tolerances.		X	
Floats or Skips	Multiple, 1/2 inch or more in combined warp and filling direction or single float or skip over more than 1 inch.			X

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TABLE III (Continued)

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Floats or Skips (Cont'd)	Multiple, less than 1/2 inch in combined warp or filling directions or single float or skip over more than 1/2 inch, but not exceeding 1 inch if in warp, or more than 1/4 inch of the width but not exceeding 1 inch, if in filling.			
Hitchback, Crack, Open Place	Clearly visible opening between adjoining picks, or warpwise tension area over part of the width resulting in visible light and heavy place.	X		
Jerked-In, Filling, Slough-Off, Slug	A clearly visible loop of filling pulled in at edges.			X
Kinks	More than 3 in any 9 linear inches.		X	
Knots	More than 1 knot in any 5 linear inches. One knot every 2 yards with untrimmed ends extending from surface of tape.		X	
Mispick, Double Pick	Two or more across the full width. Single across the full width.			X
Slack End	Two or more in the same length, jerked in between picks, or forming clearly visible loops on the surface.		X	
Slub or Slag, Gout	More than twice the thickness of the yarn (or ply, if plied).			X

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TABLE III (Continued)

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Smash	Any smash.	X		
Spot, Stain or Streak	Any clearly visible dirt, rust, grease, oil, spot, stain or streak.			X
Wrong Draw	Extending for more than 9 inches.		X	
Identification Threads	Misplaced. Missing.	 X		X

4.7.4.1.1 Yard-by-yard acceptance quality level - Zero critical and two major defects and ten total defects are permitted for every 100 one-yard units examined.

4.7.4.2 Overall examination - Examine required samples for finish, odor and cleanliness. The finish must be equal to approved sample (when available) (see 6.3). There must not be any objectionable odor and the samples must be clean.

4.7.4.3 Examination of preparation for delivery requirements - An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The Inspection Level shall be S-2 and AQL shall be 4.0 defects per 100 units.

EXAMINE	DEFECT
Marking	Omitted, illegible, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components. Bulging or distortion of container.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking - The tape shall be installed on a reel as shown on NAEL(SI) drawing 508327.

5.1.1 Reel - The reel shall be manufactured as shown on NAEL(SI) drawing 508306. NAEL(SI) will assign new part numbers to drawing 508306 for lengths not specified on drawing 508327.

5.1.2 Packing and packaging - When installing tape on reel, it shall be installed over barrier material conforming to MIL-B-81374 or MIL-B-13239, Type B-2, Class 1, Grade A which has been wrapped around the periphery of the reel hub. The barrier material shall also cover the sides of the tape reel (inside of the shipping reel) with twenty-eight (28) additional inches of paper in excess of the reel side diameters to permit sealing two sides together all around the reel (see Figure 3). The additional barrier material shall permit the opening and re-sealing of the barrier paper if examination of or work on the tape is required. Sealing process for barrier material shall not injure the tape. After installing the tape on reel and covering the barrier material, wooden cleats shall be installed around edge of reel, as shown on NAEL(SI) Drawing 508327.

5.1.3 Marking - Each reel shall be durably and legibly marked as follows:

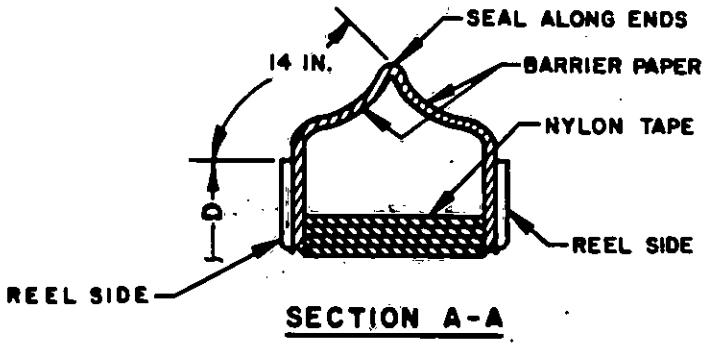
TAPE, TEXTILE, NYLON, FOR LANDBASED ARRESTING GEAR, 11-INCH
NAEL(SI) Part No.
Stock No.
Serial No.
Manufacturer's Identification Number for the Material
Contract or Order No.
Date of Manufacture
Name of Contractor
Name of Manufacturer (if not the same as contractor)

In addition, each reel shall be marked for shipment in accordance with the MIL-STD-129.

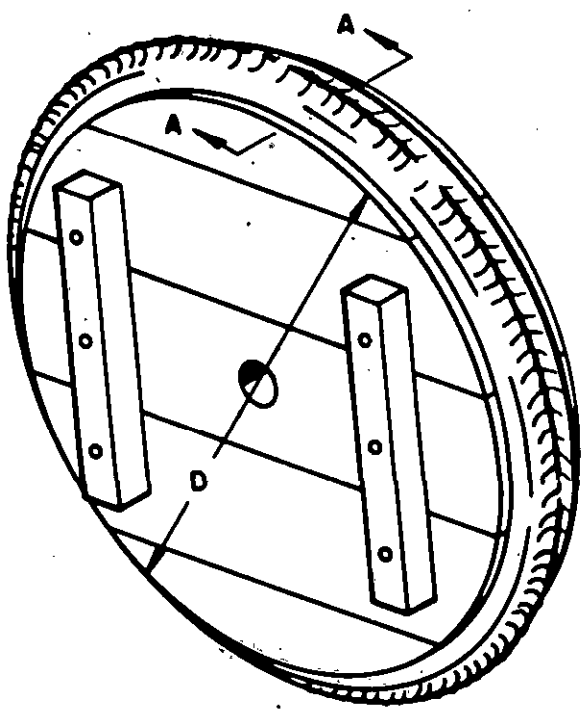
5.2 Inspection of packaged roll of tape -

5.2.1 General - The method of packing and all material entering into the process shall be subject to inspection by authorized Government Inspectors. When inspection is conducted at the contractor's plant, it shall be conducted by the contractor under the supervision of the Inspector.

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SECTION A-A



D = DIA OF REEL SIDE

Figure 3

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5.2.2 Previous approval - Acceptance of approval of material in course of packing shall in no case be construed as a guaranty of the acceptance of the finished package.

5.2.3 Examination of package - Each package shall be carefully examined to determine conformance with the requirements of this specification.

5.2.4 *Rejection and retest - Failure of any package to conform to any* one of the requirements of this specification shall be cause for rejection. Packages which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government quality assurance representative.

6. NOTES

6.1 *Intended use - The nylon tape covered by this specification is* intended for use in landbased aircraft arresting gear.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Size.

(c) Quantity desired.

(d) Manufacturer's identification.

(e) Color required *if other than specified in 3.3.*

(f) Length of roll required.

(g) Provisions for deductions of defects (if any).

(h) Responsibility for inspection records (see 4.1). Specify whether copies of records are to be furnished Government.

6.3 *For access to the standard sample, address the procuring activity* issuing the invitation for bids.

Custodian:
Navy - AS

Preparing activity:
Navy - AS
DOD Project No. 8315-N139

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81791 (AS) TAPE, TEXTILE, NYLON, LANDBASED ARRESTING GEAR, 11-INCH

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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